

1 (a) Define the radian.

.....
..... [1]

- (b) Three children are playing on a roundabout (rotatable horizontal platform) at a playground. Child X is standing on the edge of the roundabout. This child is pushing so that the edge of the roundabout moves in a circle at a constant speed of 2.4 m s^{-1} , as shown in the view from above in Figure 1.1.

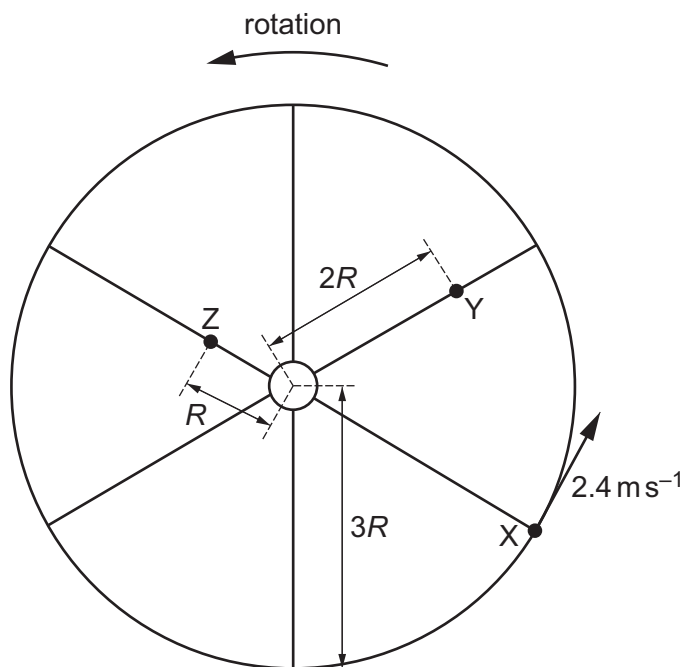


Figure 1.1

The radius of the roundabout is $3R$.

Child Y is sitting on the roundabout at a distance $2R$ from the centre.

Child Z is sitting at a distance R from the centre.

Child Y is moving in a circle with an angular speed of 0.75 rad s^{-1} .

Calculate:

- (i) the radius of the roundabout

radius = m [2]



(ii) the frequency of rotation of the roundabout

frequency = Hz [2]

(iii) the centripetal acceleration of child Y.

acceleration = ms⁻² [2]

(c) For the roundabout in **1(b)**, complete Table 1.1 to compare the linear and angular velocities for child X and child Z at any instant in time. Indicate your response by placing one tick (✓) in the appropriate box for each of magnitude and direction for the two quantities.

Table 1.1

	magnitude			direction	
	X less than Z	equal	X greater than Z	same	opposite
linear velocity					
angular velocity					

[3]